

A generator that outlives its cause: sodium channel remodelling and ectopic afferent discharge in hard flaccid syndrome

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Fourth in a series. Companion papers: *The sacral splanchnic bypass*; *Initiation without maintenance*; *The intraganglionic loop*.

Abstract

Background. Hard flaccid syndrome (HFS) typically follows a mechanical injury to the penis, yet persists for years. Interventions at the injury site — pudendal decompression, pudendal blockade — are reported to fail. Any mechanism proposed to sustain the condition must therefore be one that continues operating after the initiating lesion has healed and independently of local conditions at that site.

Hypothesis. We propose that injured axons of the dorsal nerve of the penis, and their uninjured neighbours, remodel their voltage-gated sodium channel expression, producing a persistent ectopic discharge generator distributed along the axon, in any neuroma, and within the dorsal root ganglion. On this account the generator has become anatomically dissociated from the original lesion, which is why interventions aimed at that lesion fail.

Basis. This is the best-evidenced peripheral mechanism in the neuropathic pain literature. Channel expression changes are phase-dependent: after nerve injury Nav1.7 is up-regulated, while in the acute phase Nav1.8 is down-regulated in injured neurons and up-regulated in neighbouring uninjured ones, with up-regulation extending to injured neurons in the chronic phase and maintaining the ectopic discharge. Human microneurography has recorded spontaneous activity in mechano-insensitive C-nociceptors in patients with peripheral neuropathic pain but not in healthy subjects, and the proportion of spontaneously active fibres distinguishes painful from painless polyneuropathy.

An underappreciated strength, and a serious weakness. The hypothesis accounts for something usually assigned elsewhere: partial axonal loss produces the numbness reported by 74.8% of surveyed patients, while hyperexcitability in surviving and neighbouring axons produces the paresthesia and dysesthesia — one lesion, both signs. Against that, the translational record is poor. Selective Nav1.7 blockers have repeatedly failed in neuropathic pain trials; the first approved selective Nav1.8 inhibitor is licensed for acute pain only; a follow-on Nav1.8 compound failed phase 2 in 2025; and a microneurography study reported that lidocaine did not suppress spontaneous C-nociceptor activity in painful diabetic neuropathy. The mechanism may be real and still not be druggable, and a drug challenge is therefore a poor test of it.

Tests. We propose a sequential anaesthetic localisation ladder — topical, distal infiltration, dorsal penile nerve block, pudendal block — as the primary test, because it localises the generator anatomically without depending on any drug being effective at a target. We also propose penile skin biopsy for intraepidermal nerve fibre density, and quantitative sensory testing mapped against the dorsal penile nerve distribution. None requires new technology.

Conclusion. This conjecture is the most conservative in the series: it invokes no novel biology and imports a mechanism already established in other neuropathies. Its limitation is equally clear — it explains the sensory syndrome and says nothing about the smooth muscle sign that gives the condition its name.

Keywords: hard flaccid syndrome; ectopic discharge; voltage-gated sodium channels; Nav1.7; Nav1.8; dorsal nerve of the penis; microneurography; small fibre neuropathy.

Reader advisory

This is hypothesis generation, not a report of new data. No component of the mechanism proposed here has been demonstrated in a person with hard flaccid syndrome. Sodium channel blocking agents are discussed as research probes only; intravenous lidocaine, mexiletine and related drugs carry cardiac and neurological risks and are not proposed as treatment for this condition. The anaesthetic and biopsy procedures described are research procedures requiring formal ethical approval.

1. Introduction

Hard flaccid syndrome (HFS) begins, in most reported cases, with a specific mechanical event: vigorous masturbation, penile elongation practice, or trauma during intercourse [1, 2]. It then persists. Symptom resolution was documented in only 2 of the 10 patients whose outcomes were reported in the 2025 systematic review [B] [2], and the mean age of surveyed patients was 27.4 years [B] [3] — a young population with a durable disorder.

A stretch injury to a peripheral nerve does not generate an acute afferent barrage for years. If the sensory symptoms of HFS are peripherally generated, the generator must have changed character: it must now be producing impulses on its own account rather than reporting ongoing tissue damage. There is a well-characterised way for that to happen. After nerve injury, primary afferent neurons remodel their expression of voltage-gated sodium channels, and the resulting hyperexcitability produces spontaneous ectopic discharge that outlives the injury [9, 10].

This paper states that mechanism as a hypothesis for HFS, grades the evidence, and specifies how it could be tested. It is the fourth in a series examining conjectures from a source article on HFS mechanism [5].

1.1 The role this conjecture plays, and its limits

Two things should be said at the outset, because they define what this hypothesis can and cannot be asked to do.

It supplies something the central hypothesis needs. Central sensitization is characterised as reversible in principle [A] [23]; a condition that almost never remits therefore requires something continuing to feed it [7]. A remodelled ectopic generator is a candidate for that driver. On this reading the present hypothesis is not a rival to the central account but a component of it.

It does not explain the hard flaccid state. An ectopic afferent generator produces afferent traffic. Converting that into sustained cavernosal contraction requires the somato-autonomic reflex proposed elsewhere in this series [6, 7]. Presented alone, this hypothesis is an account of the sensory syndrome and nothing more. We think it should be presented that way, and we resist the temptation — available at several points below — to claim more.

2. Methods

This is a hypothesis and theory article. No new human or animal data were generated; no patients were involved; ethical approval was not required.

Literature was identified by targeted searching of MEDLINE/PubMed and the open web between May and August 2026, using combinations of the terms *ectopic discharge*, *voltage-gated sodium channel neuropathic pain*, *Nav1.7*, *Nav1.8*, *microneurography C-nociceptor*, *intraepidermal nerve fibre density*, *dorsal nerve of the penis* and *hard flaccid syndrome*, with backward citation chasing. The search was purposive and non-systematic; no protocol was registered and no formal risk-of-bias appraisal was applied. We searched specifically for the clinical translation record of sodium channel blockade, and Section 5 reports what that returned.

Claims are graded as in Table 1. Claims not verified against a retrievable source are marked [VERIFY]; quantities that do not exist in the literature are marked [DATA NEEDED].

Table 1. Evidence grading scheme. The grade refers to the strength of evidence for the *stated proposition*, not to its relevance to hard flaccid syndrome. No proposition here carries grade A evidence *in HFS*, because none exists.

Grade	Definition
[A]	Replicated human evidence: consistent findings across independent human studies.
[B]	Single human study, human reference text, congress abstract, or human data with substantial methodological limitation.
[C]	Experimental animal evidence, with species and preparation stated.
[D]	Mechanistic inference: deduced from A–C evidence but not itself directly observed.
[E]	Unverified anecdotal or community-sourced report; not peer reviewed, denominators unknown.

3. Background

3.1 Channel remodelling after nerve injury is phase-dependent

The relevant biology is unusually well specified. Nerve injury increases the activity of kinases including p38 MAPK and ERK1/2, leading to modulation and up-regulation of Nav1.7, which contributes to increased generation of ectopic discharge. In the acute phase after nerve damage, Nav1.8 is down-regulated in injured neurons and up-regulated in neighbouring uninjured neurons, contributing to increased spontaneous discharge; in the chronic phase, through crosstalk between injured and intact neurons, Nav1.8 is up-regulated in injured neurons as well, further increasing and maintaining

the ectopic discharge [C] [9].

Three features of that account matter for HFS [D]:

1. **Two populations are involved, not one.** Uninjured neighbours become hyperexcitable. A partial injury therefore recruits more neurons into the abnormal state than it damages.
2. **The mechanism is explicitly a maintenance mechanism.** The chronic-phase change is described as maintaining the ectopic discharge, which is precisely the property a hypothesis of HFS chronicity requires.
3. **The generator is not at the lesion.** Ectopic activity in nerve-end neuroma, compressed nerves or roots, and dorsal root ganglia may underlie spontaneous pain in different conditions [A] [10]. A generator distributed along the axon and within the ganglion is not removed by releasing a compression point.

The cellular mechanism of the discharge itself has been characterised: burst discharge in primary sensory neurons is triggered by subthreshold membrane potential oscillations and maintained by depolarising afterpotentials [C] [VERIFY] [11].

3.2 Human evidence that this happens in people

The animal literature would be weak grounds on its own. It is not the only evidence.

Human genetics anchors the pathway: gain-of-function mutations in *SCN9A*, encoding Nav1.7, cause inherited erythromelalgia, characterised by repeated episodes of intense pain, through mechanisms including a hyperpolarising shift in channel activation that renders nociceptors hyperexcitable [A] [9].

Human microneurography provides the direct functional evidence. In a study comparing patients with peripheral neuropathic pain ($n=20$), healthy subjects ($n=15$) and five rat models, spontaneous activity was present in a proportion of mechano-insensitive C-nociceptors in the patients and in all rat models; the patients showed an intermediate proportion (33.3%) between focal traumatic nerve injury models (59.5%) and polyneuropathy models (18.6%), and spontaneously active mechano-sensitive C-nociceptors were not recorded [B] [12]. In a separate study of 214 C-nociceptors from patients with painful and painless polyneuropathy, patients with and without pain were distinguished by the occurrence of spontaneous activity and mechanical sensitisation in C-nociceptors, and hyperexcitability in mechano-insensitive C-nociceptors was significantly higher in the painful group [B] [13]. Pathological C-fibre behaviour has been documented in patients with a chronic painful condition [B] [VERIFY] [14].

This is a stronger human evidence base than any other conjecture in this series enjoys. It is also entirely from other conditions.

3.3 One lesion, both sensory signs

The single most commonly reported sensory symptom in HFS is *numbness* — loss of sensation somewhere on the penis in 74.8% of surveyed patients [B] [3]. Numbness is a negative sensory sign, and an ectopic generator produces positive signs. This looks at first like a problem for the hypothesis, and in the source article the sensory loss is assigned to a different conjecture concerning partial deafferentation [5].

We think this hypothesis accounts for both, and that this has been underappreciated. A partial stretch injury to the dorsal nerve of the penis destroys some axons and spares others. The destroyed axons produce reduced afferent transmission — numbness, and loss of temperature, pressure, vibration and texture discrimination. The surviving injured axons and their uninjured neighbours undergo the channel remodelling described above and generate ectopic discharge — paresthesia and dysesthesia. **The coexistence of numbness and abnormal sensation in the same territory is not a contradiction but the expected signature of a partial nerve lesion [D]**, and the two-population structure of the channel biology [9] supplies the mechanism directly.

We flag this as an inference rather than a finding, and note that it is not unique to this hypothesis: the same coexistence is predicted by partial deafferentation with central hyperexcitability. What is specific here is that the positive signs are attributed to a peripheral generator rather than to central gain — and that difference is testable (Section 6).

3.4 Which nerve, and where

The dorsal nerve of the penis is the terminal sensory branch of the pudendal nerve; its peripheral distribution in humans has been mapped [B] [VERIFY] [28], and the glans is densely supplied by free nerve endings [B] [VERIFY] [29]. A stretch injury sustained during vigorous manual practice would be expected to act on the nerve over its course along the dorsum of the shaft, not at a single compression point. Under this hypothesis the resulting generator would be distributed along that course, at any neuroma or injury site, and in the corresponding sacral dorsal root ganglia [D].

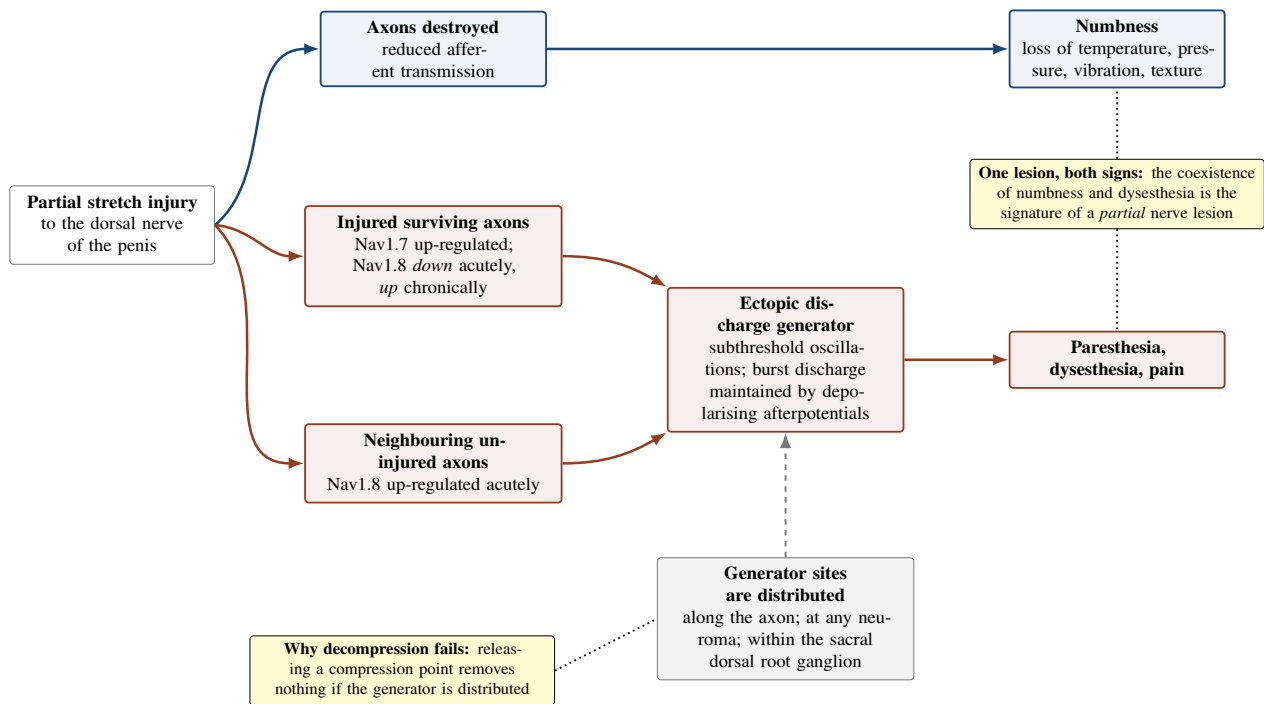


Figure 1. Proposed consequences of a partial stretch injury to the dorsal nerve of the penis. A partial lesion divides the afferent population into three groups. Destroyed axons produce negative sensory signs. Surviving injured axons and their uninjured neighbours undergo phase-dependent voltage-gated sodium channel remodelling and become the source of ectopic discharge, producing positive sensory signs. Because the resulting generator is distributed along the axon, at any neuroma, and within the dorsal root ganglion rather than being located at a single compression point, interventions directed at the original injury site would be expected to fail. **Note that this diagram accounts for the sensory syndrome only; it says nothing about cavernosal tone.** Channel expression changes shown are from rodent nerve injury models; the schema is an inference, not an observation in this condition.

4. The hypothesis

Formal statement

H4 (ectopic afferent generator from channel remodelling). In hard flaccid syndrome, injured axons of the dorsal nerve of the penis and their uninjured neighbours remodel their voltage-gated sodium channel expression, producing a persistent ectopic discharge generator that is anatomically distributed — along the axon, at any injury site or neuroma, and within the corresponding dorsal root ganglia — and that continues to fire independently of local conditions at the original lesion.

H4-peripheral (the strong form). The generator is accessible distally. Prediction: local anaesthetic applied peripheral to the ganglion abolishes the positive sensory symptoms.

H4-proximal. The generator lies principally in the dorsal root ganglion or proximal axon. Prediction: distal anaesthesia does not abolish symptoms; proximal blockade does.

H4-driver (the modest form). The generator exists but its main significance is as the input that sustains a central process. Prediction: silencing it reduces symptoms only partially and only while it is silenced.

H4-null. The positive sensory symptoms are centrally generated; no peripheral generator is required.

Auxiliary assumptions:

- **A1 (injury sufficiency).** A stretch injury sustained during masturbation is sufficient to trigger channel remodelling. The animal literature uses ligation, transection, crush and chronic constriction; stretch injury of this kind has not been modelled [DATA NEEDED].
- **A2 (territory).** Remodelling occurs in the dorsal penile nerve and sacral ganglia. Never examined.
- **A3 (afferent-to-autonomic transfer).** For H4 to bear on the hard flaccid state at all, the ectopic afferent traffic must engage a somato-autonomic reflex [7]. Without A3, H4 explains sensation only — and we regard A3 as belonging to the companion hypothesis rather than to this one.

5. The translation problem

This hypothesis has the strongest mechanistic evidence base in the series and the worst therapeutic track record. Both facts are important, and the second is usually omitted.

1. **Selective Nav1.7 blockers have repeatedly failed.** PF-05089771, a highly selective Nav1.7 inhibitor, failed to meet efficacy criteria in painful diabetic peripheral neuropathy [B] [VERIFY] [16, 17]. Vixotrigine has been trialled in trigeminal neuralgia and small fibre neuropathy with mixed and largely disappointing results [B] [VERIFY] [18, 16]. A general assessment of the field concluded that despite intensive preclinical work, clinical tests of Nav1.7 blockers have yielded disappointing results [B] [VERIFY] [17].
2. **The one approved selective agent is licensed for acute pain only.** Suzetrigine (VX-548), a selective Nav1.8 inhibitor, was approved by the US Food and Drug Administration on 30 January 2025 for moderate-to-severe acute pain; trials in chronic and neuropathic pain were ongoing at that time [B] [VERIFY] [19]. It is not a validated probe of neuropathic mechanisms.
3. **A follow-on compound failed.** VX-993, a further selective Nav1.8 inhibitor, did not outperform placebo in a phase 2 acute pain study reported in August 2025 and was discontinued as monotherapy [B] [VERIFY: TRADE PRESS AND COMPANY ANNOUNCEMENT; NO PEER-REVIEWED REPORT IDENTIFIED] [20].
4. **Most damagingly for the obvious test.** A microneurography study reported that lidocaine had no impact on the spontaneous activity of peripheral C-nociceptors in patients with painful diabetic neuropathy [B] [VERIFY] [21]. If a sodium channel blocking local anaesthetic does not silence the generator it is supposed to silence, then a negative drug challenge in HFS would not refute H4 — and a positive one would be hard to attribute.

The correct inference is not that the mechanism is false. Drug failure has many causes: insufficient target engagement at tolerated doses, redundancy across channel subtypes, and heterogeneity in the trial populations. The correct inference is narrower and more useful: **a pharmacological challenge is a poor test of this hypothesis, and the test should be anatomical instead.**

6. Predictions and tests

6.1 Primary test: the anaesthetic localisation ladder

The distinguishing question is not whether abnormal discharge exists but *where it is generated*. That is answerable by graded, anatomically sequential local anaesthesia, and the logic does not depend on any drug being effective at a molecular target — only on local anaesthetic doing what local anaesthetic reliably does, which is block conduction where it is applied.

The relevant precedent is explicit in the literature: in some patients with post-traumatic pain, local anaesthesia of the relevant peripheral nerve does not alleviate the pain, implying that ectopic impulses arise either centrally or in proximal parts of the primary afferent neurons [B] [22]. Conversely, complete relief of polyneuropathy-related pain after an ultrasound-guided peripheral nerve block with lidocaine has been reported in an open-label study, supporting a peripheral generator in that population [B] [VERIFY] [10].

Prediction P1. Positive sensory symptoms will be abolished by anaesthesia applied at or proximal to the generator, and unaffected by anaesthesia applied distal to it, with the transition identifying the site.

Design requirements: separate sessions; vehicle or sham control at each level; blinded outcome assessment; documentation that each block was technically adequate by demonstrated cutaneous anaesthesia in the expected territory; and a pre-specified sensory outcome measure rather than global impression. The single most important design point is that **failure of a block must be distinguished from failure to achieve a block** — the same methodological requirement identified in both companion papers [6, 7].

6.2 Secondary test: penile skin biopsy for intraepidermal nerve fibre density

If the initiating lesion is partial axonal loss (Section 3.3), it should be visible. Skin punch biopsy with PGP 9.5 immunostaining and quantification of intraepidermal nerve fibre density is the established, guideline-endorsed method for demonstrating small fibre loss, with distal-leg biopsy carrying a Level A recommendation [A] [VERIFY] [26, 27].

Prediction P2. Intraepidermal nerve fibre density in penile shaft skin will be reduced in men with HFS relative to matched controls, and the reduction will correlate with the extent of the numb territory.

The obstacle is normative data. Published reference values are for distal leg, thigh and forearm [A] [VERIFY] [26]; we identified no normative dataset for penile or genital skin [DATA NEEDED]. Establishing age-stratified genital reference values in healthy volunteers is a prerequisite, and is a self-contained study that would be useful well beyond this hypothesis. Acceptability of genital skin biopsy in this population is also unknown and should be assessed before a

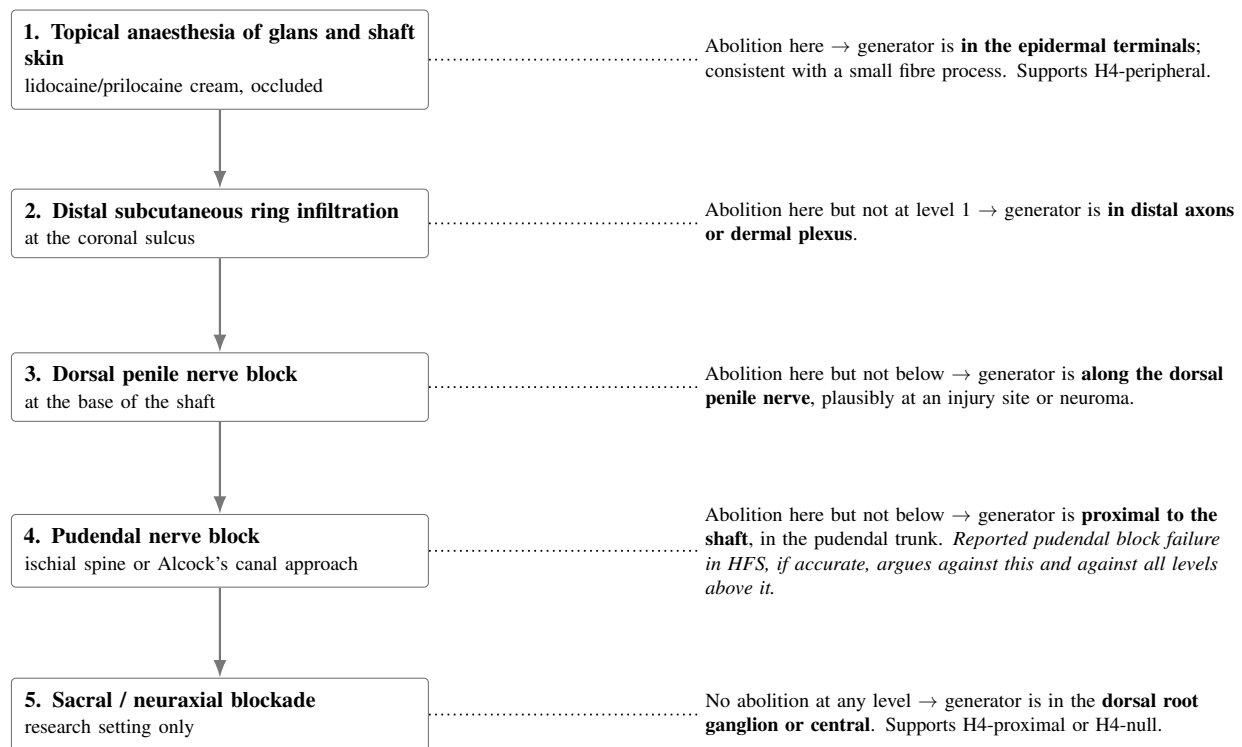


Figure 2. The anaesthetic localisation ladder. Sequential local anaesthesia, applied from distal to proximal on separate occasions with blinded symptom assessment and vehicle control, localises an ectopic generator anatomically. The logic does not depend on any drug engaging a molecular target — only on conduction block occurring where anaesthetic is applied — which is why it is preferred here over a sodium-channel-blocker challenge (Section 5). Each level must be documented as technically adequate by demonstrated cutaneous anaesthesia in the expected territory. All procedures shown are research procedures requiring ethical approval; levels 4 and 5 are invasive and would need separate justification.

study is designed.

6.3 Tertiary test: quantitative sensory testing with territorial mapping

Prediction P3. Quantitative sensory testing will show stimulus-independent paresthesia and dysesthesia together with elevated detection thresholds *within* the distribution of the dorsal nerve of the penis [28], with abnormality respecting that distribution rather than crossing it.

The discriminating value lies in the boundary. A deficit confined to the nerve territory supports a peripheral lesion; extraterritorial spread supports central amplification [7]. A standardised, multicentre-validated battery with published reliability exists [A] [24, 25], though again the reference values are for face, hand and foot and genital control data would be required [DATA NEEDED].

6.4 The direct test, and why it is probably not available

Microneurography of the dorsal nerve of the penis, seeking spontaneous activity in mechano-insensitive C-nociceptors as has been demonstrated in other neuropathic populations [12, 13], would be the decisive test. We identified no report of microneurography performed on this nerve; feasibility, tolerability and yield are unknown [DATA NEEDED]. It is technically demanding even in accessible limb nerves and requires specialised expertise available in few centres. We note it for completeness and do not propose it as a near-term study.

6.5 A pharmacological prediction, offered with reservations

Prediction P4. A sodium channel blocking agent at adequate dose reduces paresthesia and dysesthesia while leaving the hard flaccid state unchanged.

The reported clinical observation — that gabapentin helps pain but not the hard flaccid state [E] [5] — is consistent with this shape of result, though gabapentinoids act at the $\alpha 2\delta$ -1 subunit rather than on sodium channels and therefore dampen the consequences of the barrage rather than abolishing its source [D]. Given the translation record in Section 5, and the specific finding that lidocaine did not suppress spontaneous C-nociceptor activity in painful diabetic neuropathy [21], we regard P4 as weakly informative in either direction and would not design a study around it.

Table 2. Predictions of H4 compared with the principal alternatives in this series.

Observation	H4 (ectopic generator)	Central sensitization	Ganglionic sympathetic coupling	Discriminating value
Symptoms abolished by anaesthesia distal to the ganglion	Predicted (peripheral form)	Not predicted	Not predicted	High
Sensory abnormality respects the dorsal penile nerve territory	Predicted	Not predicted (spread expected)	Not predicted (diffusion spread)	High
Reduced intraepidermal nerve fibre density in penile skin	Predicted	Not predicted	Not predicted	Moderate–high
Coexisting numbness and dysesthesia	Predicted	Predicted	Not predicted (does not address numbness)	Low
Spontaneous activity in C-nociceptors on microneurography	Predicted	Not predicted	Possible	Decisive but probably unavailable
Symptoms vary with sympathetic arousal	Not predicted	Not specifically predicted	Predicted	High, against H4
Failure of pudendal decompression and blockade	Predicted (proximal form)	Predicted	Predicted	None
Sodium channel blocker reduces positive symptoms only	Predicted	Not predicted	Not predicted	Low (see Section 5)

7. Alternative explanations

1. **Central generation.** Positive sensory symptoms in a partially deafferented territory can be generated centrally without any peripheral generator [23, 7]. This is H4's principal competitor, and the localisation ladder is designed specifically to separate them.
2. **Sympathetic–sensory coupling at the ganglion.** If sensory discharge is driven by sympathetic activity rather than arising autonomously, the symptom pattern should track sympathetic arousal [8]. The reported worsening with masturbation in 75.9% [3] fits that account better than it fits H4, which predicts symptoms largely independent of autonomic state. **We regard this as the strongest single argument against H4 as a complete sensory account.**
3. **Mechanical rather than neural.** Symptoms provoked by manual activity may reflect mechanical irritation of an injured nerve rather than any spontaneous generator. This is easily tested by comparing symptom change after orgasm achieved with and without direct penile stimulation [DATA NEEDED].
4. **No lesion at all.** Only 58.0% of surveyed patients reported onset after a specific incident [B] [3]. H4 requires a nerve injury and can absorb the remaining 42.0% only by positing a subclinical one.
5. **Attentional amplification.** Depression or anxiety was reported in 75% of the largest series [B] [4, 2]. Sustained hypervigilant attention to genital sensation is itself capable of producing and maintaining symptom reports, and is not distinguished from ectopic discharge by any self-report measure. Blinded threshold testing, not symptom rating, is the way to separate them.

8. Implications

No clinical implications are drawn. Three research implications follow.

First, localise before pharmacologising. The field's instinct is to trial drugs. For this hypothesis that is the wrong order: the anatomical question is answerable with local anaesthetic and blinded assessment, and answering it would tell you which drugs could possibly work.

Second, generate genital normative data. Both the biopsy test and the sensory testing protocol are blocked by the same gap: no reference values for genital skin. A single study in healthy volunteers would unblock several lines of investigation at once and is not conceptually difficult.

Third, record whether abnormality respects the nerve territory. Whether sensory abnormality is confined to the dorsal penile nerve distribution is the cheapest high-value observation in this whole series, and distinguishes peripheral from central accounts across three of the four papers.

9. Limitations

1. **No evidence in HFS.** Channel remodelling, ectopic discharge and axonal loss have never been sought in this condition.
2. **Wrong injury model.** The supporting animal literature uses ligation, transection, crush and constriction. Stretch injury of the kind implicated in HFS has not been modelled.

3. **Wrong nerve, wrong ganglion.** Nothing in the sodium channel literature concerns the dorsal penile nerve or the sacral ganglia.
4. **It does not explain the defining sign.** H4 is an account of sensation. Presented as an explanation of hard flaccid syndrome it would be overreaching, and we have tried to avoid doing so.
5. **It fits the arousal-dependence poorly.** The most striking symptom correlate in the survey data — worsening with masturbation in 75.9% [3] — is better explained by a hypothesis this one competes with.
6. **The translation record is poor,** and the most obvious pharmacological test is compromised by evidence that lidocaine does not suppress the activity in question [21].
7. **The primary test has real burdens.** The localisation ladder requires multiple sessions, invasive blocks at the upper levels, and a population in which genital procedures may be poorly tolerated. Its acceptability is unknown [DATA NEEDED].
8. **Both secondary tests are blocked by missing normative data** for genital skin.
9. **Non-systematic search,** with several citations verified only against secondary sources; all are marked [VERIFY]. Two citations in Section 5 rest on trade press and company announcements rather than peer-reviewed reports and are flagged as such.
10. **Confirmation risk.** This paper was written to develop a specific conjecture. We have tried to counter it by reporting the translation failures in full, by identifying the observation that fits a rival hypothesis better, and by declining to design a study around the prediction most likely to produce a spurious positive. The reader should still discount accordingly.

10. Refutation criteria

- **Substantially refuted (peripheral form).** No level of the anaesthetic ladder, up to and including a documented pudendal block, abolishes positive sensory symptoms.
- **Substantially refuted.** Quantitative sensory testing shows sensory abnormality extending well beyond the distribution of the dorsal nerve of the penis, with no territorial boundary.
- **Weakened.** Penile intraepidermal nerve fibre density is normal in men with HFS against appropriate genital reference values.
- **Weakened.** Symptom intensity is shown to track sympathetic arousal closely, with a latency and magnitude better explained by sympathetic–sensory coupling [8].
- **Weakened.** Symptom worsening with masturbation is shown to depend on mechanical stimulation rather than on any autonomous discharge.
- **Rendered moot.** Microneurography of the dorsal penile nerve, if it ever becomes feasible, shows no spontaneous activity in mechano-insensitive C-nociceptors.

11. Conclusion

Of the conjectures examined in this series, this is the most conservative. It invokes no novel biology. It imports a mechanism that is established in human genetics, demonstrated by microneurography in human patients, and characterised at the level of individual channel subtypes and their phase-dependent expression. It explains why interventions at the original injury site fail, and — a point we think has been underappreciated — it explains the coexistence of numbness and dysesthesia in the same territory as the ordinary signature of a partial nerve lesion rather than as a paradox requiring two mechanisms.

Its weaknesses are equally clear and we have not softened them. It says nothing about the sign the condition is named for. It fits poorly with the most striking symptom correlate in the survey data, the worsening with sexual activity, which a competing hypothesis explains better. And the field it borrows from has an unhappy therapeutic history: selective Nav1.7 blockers have failed repeatedly, the one approved selective Nav1.8 agent is licensed for acute pain only, a follow-on compound failed in 2025, and lidocaine has been reported not to silence the very activity this hypothesis proposes.

That last fact changes what the right experiment is. Because a drug challenge is compromised, the test should be anatomical: a graded, blinded, vehicle-controlled anaesthetic ladder from glans skin to pudendal trunk, with each block documented as adequate. It answers a question no drug trial can answer — where the impulses are being made — and the answer would constrain not only this hypothesis but the three that accompany it. It requires no new technology and no new molecule. It requires only that someone record what happens, and at which level.

Declarations

Authorship and the use of artificial intelligence. The second listed author is a large language model. It is listed at the corresponding author's request and in the interest of transparency, but this contravenes the recommendations of the International Committee of Medical Journal Editors, which hold that artificial intelligence tools cannot meet authorship criteria because they cannot take responsibility for the accuracy and integrity of the work. **Before submission to any peer-reviewed venue this attribution should be converted into an AI-use disclosure, with the human author accepting sole responsibility for the content.** Every citation, and particularly those marked [VERIFY], should be checked against the primary source first; the two references in Section 5 sourced from trade press should be replaced with peer-reviewed reports or removed.

Ethics approval. Not applicable. No human participants, animals or identifiable data were involved.

Funding. None.

Competing interests. No competing financial interests. The first author maintains a public website on hard flaccid syndrome and authored the source article [5]; this is a non-financial interest in the hypothesis being taken seriously, and readers should weigh the argument accordingly.

Data availability. No datasets were generated or analysed.

Clinical disclaimer. This article is hypothesis generation. It does not constitute medical advice and does not establish a clinician–patient relationship. Nothing in it should be used to select, request or avoid any treatment. Sodium channel blocking agents including intravenous lidocaine and mexiletine carry cardiac and neurological risks; nerve blocks and skin biopsy are invasive procedures. All are appropriate only under qualified medical supervision and, in this condition, only within an approved research protocol.

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